

DIFFERENTIAL EQUATIONS

Lecture 2 — Definitions: Order, Degree, Linearity & Solutions

Differential Equation

Definition: Differential Equation

An equation involving a **dependent variable** and its **derivative(s)** with respect to one or more independent variables.

The dependent variable is the one being differentiated (typically y), and the independent variable is the one we differentiate with respect to (typically x or t).

ODE and PDE

Definition: Ordinary Differential Equation (ODE)

A DE containing **ordinary derivatives** of the dependent variable with respect to **exactly one** independent variable.

Definition: Partial Differential Equation (PDE)

A DE containing **partial derivatives** of the dependent variable with respect to **more than one** independent variable.

Order and Degree

Definition: Order

The **order** of a DE is the order of the **highest derivative** that appears in the equation.

Definition: Degree

The **degree** of a DE is the **greatest exponent (power)** of the highest-order derivative, after the equation has been cleared of all **radicals and fractions** involving derivatives.

Important: Before reading off the degree, you must ensure:

1. No derivative appears under a radical or with a fractional exponent — raise both sides to the appropriate power to clear it.
2. No derivative appears in a denominator — multiply through to clear it.
3. If any derivative appears inside a transcendental function such as $\sin$, $e^{(\cdot)}$, or $\ln$, the degree is **undefined**.

These conditions apply to **every** derivative in the equation, not just the highest-order one.

Worked Examples — Order and Degree

Example 1: $\frac{dy}{dx} + y \cos x = \sin x$

The highest derivative is dy/dx — order **1**. It appears to the power 1 — degree **1**. This is an **ODE**.

$$\text{Order} = 1, \quad \text{Degree} = 1$$

Example 2: $\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0$

The highest derivative is d^2y/dx^2 — order **2**. It appears to the power 1 — degree **1**. This is an **ODE**.

$$\text{Order} = 2, \quad \text{Degree} = 1$$

Note: the $(dy/dx)^2$ term does not affect the degree because d^2y/dx^2 is the *highest* derivative, and it appears to power 1.

Example 3: $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2} = \frac{d^2y}{dx^2}$

The highest derivative is d^2y/dx^2 but first we must clear the fractional power on the LHS. Square both sides:

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = \left(\frac{d^2y}{dx^2}\right)^2$$

Now the highest derivative d^2y/dx^2 appears to the power **2**.

$$\text{Order} = 2, \quad \text{Degree} = 2$$

Example 4: $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$

Contains partial derivatives with respect to both x and y — this is a **PDE**. The highest partial derivative is first order, appearing to the power 1.

$$\text{Order} = 1, \quad \text{Degree} = 1$$

Example 5: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$ (Laplace's Equation)

Contains second-order partial derivatives with respect to x , y , and z — a **PDE**. Each appears to the power 1.

$$\text{Order} = 2, \quad \text{Degree} = 1$$

Order and Degree Summary Table

No.	Equation	Type	Order	Degree
1	$dy/dx + y \cos x = \sin x$	ODE	1	1
2	$d^2y/dx^2 + xy(dy/dx)^2 = 0$	ODE	2	1
3	$[1 + (dy/dx)^2]^{3/2} = d^2y/dx^2$	ODE	2	2
4	$x \partial u / \partial x + y \partial u / \partial y = nu$	PDE	1	1
5	$\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 + \partial^2 u / \partial z^2 = 0$	PDE	2	1

Linear vs Non-Linear DEs

Definition: Linear DE

A DE is **linear** if **all three** of the following conditions hold:

1. The **degree** of the DE is **1**.
2. There is **no product** of the dependent variable and/or its derivatives with each other.
3. **No transcendental function** (trigonometric, exponential, logarithmic) of the dependent variable appears.

If *any* condition is violated the DE is **non-linear**.

Key distinction: The three conditions apply only to the **dependent variable** y and its derivatives. The independent variable x may appear in any form — $x^{1/2}$, $\sin(x)$, e^x , $\ln(x)$ — without affecting linearity or degree. These are simply **variable coefficients**.

Examples of Each Condition Being Violated

Condition 1 violated — Degree $\neq 1$

$$\left(\frac{d^2y}{dx^2}\right)^2 + \frac{dy}{dx} = x \quad \Rightarrow \quad \text{Non-linear}$$

The highest derivative d^2y/dx^2 appears squared $\Rightarrow$ degree = 2 $\neq$ 1.

Condition 2 violated — Product of y and/or its derivatives

$$y \frac{dy}{dx} = x \quad \Rightarrow \quad \text{Non-linear} \quad (y \text{ multiplied by its derivative})$$

$$\frac{d^2y}{dx^2} \cdot \frac{dy}{dx} = x \quad \Rightarrow \quad \text{Non-linear} \quad (\text{product of two derivatives})$$

$$\frac{dy}{dx} + y^2 = 0 \quad \Rightarrow \quad \text{Non-linear} \quad (y^2 = y \cdot y, \text{ product of } y \text{ with itself})$$

Condition 3 violated — Transcendental function of y

$$\frac{dy}{dx} + \sin(y) = x \quad \Rightarrow \quad \text{Non-linear} \quad (\text{trig of } y)$$

$$\frac{d^2y}{dx^2} + e^y = 0 \quad \Rightarrow \quad \text{Non-linear} \quad (\text{exponential of } y)$$

$$\frac{dy}{dx} = \ln(y) \quad \Rightarrow \quad \text{Non-linear} \quad (\text{log of } y)$$

These are still Linear — x can do anything

$$\frac{dy}{dx} + \sin(x) \cdot y = e^x \quad \Rightarrow \quad \text{Linear} \checkmark \quad (\sin(x) \text{ is just a coefficient})$$

$$\frac{d^2y}{dx^2} + x^2 \frac{dy}{dx} + \ln(x) \cdot y = 0 \quad \Rightarrow \quad \text{Linear} \checkmark \quad (\text{all weird functions are of } x, \text{ not } y)$$

$$\frac{dy}{dx} + xy = x^3 \quad \Rightarrow \quad \text{Linear} \checkmark \quad (xy \text{ means coefficient } x \text{ times } y, \text{ not } y \text{ times } y)$$

Solutions of a DE

Definition: Solution (Integral) of a DE

A relation between the dependent and independent variables that:

1. Contains **no derivatives**, and
2. When substituted back into the DE, **identically satisfies** it for all values of the independent variable.

General Solution

Definition: General Solution

A solution that contains exactly n **arbitrary constants**, where n is the **order** of the DE. It represents the complete family of all possible solutions.

$$\text{Number of arbitrary constants} = \text{Order of the DE}$$

The reason for this is that solving an n th-order DE requires n integrations, and each integration introduces one arbitrary constant.

Particular Solution

Definition: Particular Solution

A solution obtained from the general solution by assigning **specific numerical values** to the arbitrary constants. These values are determined by **initial conditions** or **boundary conditions**.

Example: General vs Particular Solution of $dy/dx = \lambda y$ **Solve:** Separate variables and integrate:

$$\frac{1}{y} dy = \lambda dx \implies \ln|y| = \lambda x + C \implies y = e^{\lambda x + C} = e^C \cdot e^{\lambda x}$$

Since $e^C > 0$ always, we rename $e^C = C_1$ (now $C_1 \in \mathbb{R}$, unrestricted):

$$\textbf{General solution: } y = C_1 e^{\lambda x}$$

This contains 1 arbitrary constant C_1 , matching the 1st-order DE.**Apply IC** $y(0) = 3$:

$$3 = C_1 e^0 = C_1 \implies C_1 = 3$$

$$\textbf{Particular solution: } \boxed{y = 3e^{\lambda x}}$$

Why rename e^C as C_1 ? The constant e^C is always strictly positive (since $e^C > 0$ for all real C). This secretly restricts solutions to positive multipliers only, losing solutions like $y = -2e^{\lambda x}$. Renaming $e^C = C_1$ where $C_1 \in \mathbb{R}$ unlocks the full set of solutions including negative and zero values.

Summary Table

Concept	Definition / Key Point
Differential Equation	Equation involving a dependent variable and its derivatives
ODE	One independent variable; ordinary derivatives
PDE	More than one independent variable; partial derivatives
Order	Order of the highest derivative present
Degree	Power of the highest derivative, after clearing radicals and fractions
Undefined Degree	When a derivative appears inside $\sin$, $e^{(\cdot)}$, $\ln$, etc.
Linear DE	Degree 1, no products of y /derivatives, no transcendentals of y
General Solution	Contains n arbitrary constants (n = order of DE)
Particular Solution	General solution with constants fixed by ICs or BCs